

ASHISH KUMAR SHAW

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EDUCATION

Bachelor of Computer Applications (B.C.A.) – Cloud Security Specialization

Amity University Online, Noida, India — Expected Graduation: July 2026

Relevant Coursework: Cloud Security, Network Security (VPC), Cryptography, Backend Development

INTERNSHIP EXPERIENCE

Cloud Security Intern — *TCS iON, Remote* — June 2025 – August 2025

- Configured 8+ VPC security groups and network policies under mentor guidance for secure cloud infrastructure
- Learned and applied security best practices including least privilege access across 5+ IAM roles
- Documented 12+ security configurations and procedures for team knowledge base

KEY PROJECTS

SecureChat – E2E Encrypted Messaging Backend — kavro.duckdns.org/docs

- Built messaging backend with 8 RESTful endpoints using Python FastAPI achieving 100% client-side encryption (NaCl), Redis rate limiting, and account lockout after 5 failed attempts
- Deployed on AWS EC2 with Docker, automated Nginx and SSL setup; validated with 25 Pytest tests
- *Tech: Python, FastAPI, PostgreSQL, Redis, Docker, AWS EC2, NaCl Cryptography*

EventFlow – Event-Driven Notification Service — eventdriven.duckdns.org/docs

- Built asynchronous notification system with Redis producer-consumer pattern across 3 worker queues
- Implemented retry logic, exponential backoff, DLQ for zero data loss; integrated Prometheus metrics
- *Tech: Python, FastAPI, Redis, Docker, AWS EC2, Prometheus*

Cloud Fun Fact Generator – Serverless — cfunfact.onrender.com

- Built serverless app using AWS Lambda, API Gateway, and DynamoDB with sub-2-second response times
- *Tech: AWS Lambda, API Gateway, DynamoDB, AI Integration*

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL — **Frameworks:** FastAPI, REST APIs

Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, API Gateway, DynamoDB), Docker, Linux

Databases: PostgreSQL, MySQL, Redis, DynamoDB

Security: E2EE (NaCl), JWT Authentication, VPC Configuration, Rate Limiting, HTTPS/SSL

Tools: Git/GitHub, Prometheus, Pytest, Nginx

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations — Applied Cloud Computing (TCS iON) — Information Security (TCS iON) — Cloud Security Fundamentals (Simplilearn)

All certificates: drive.google.com/certificates